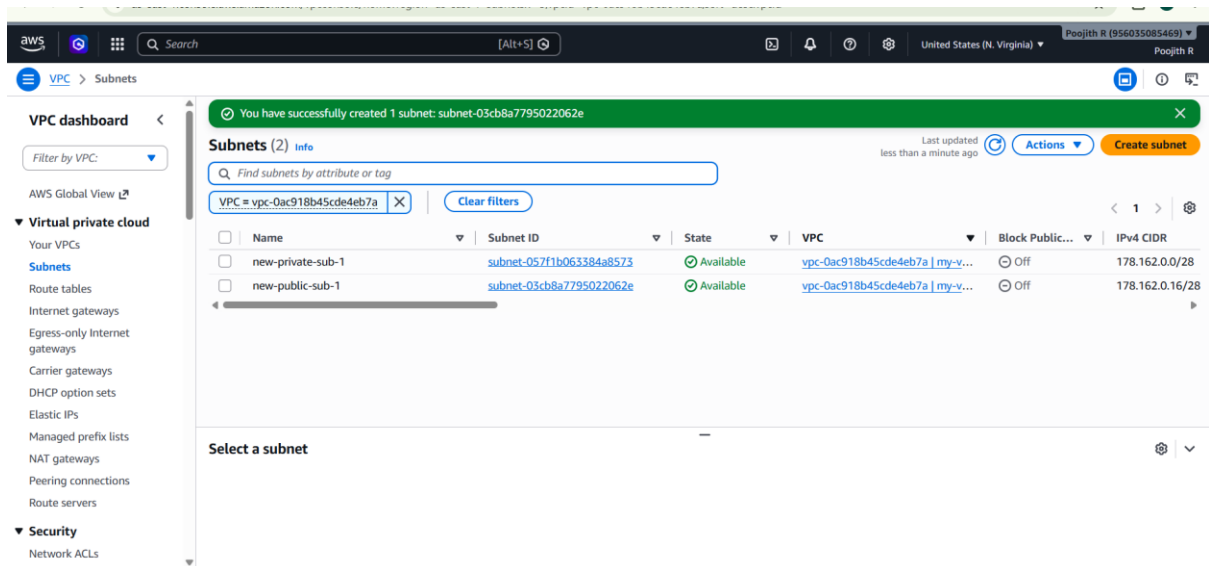
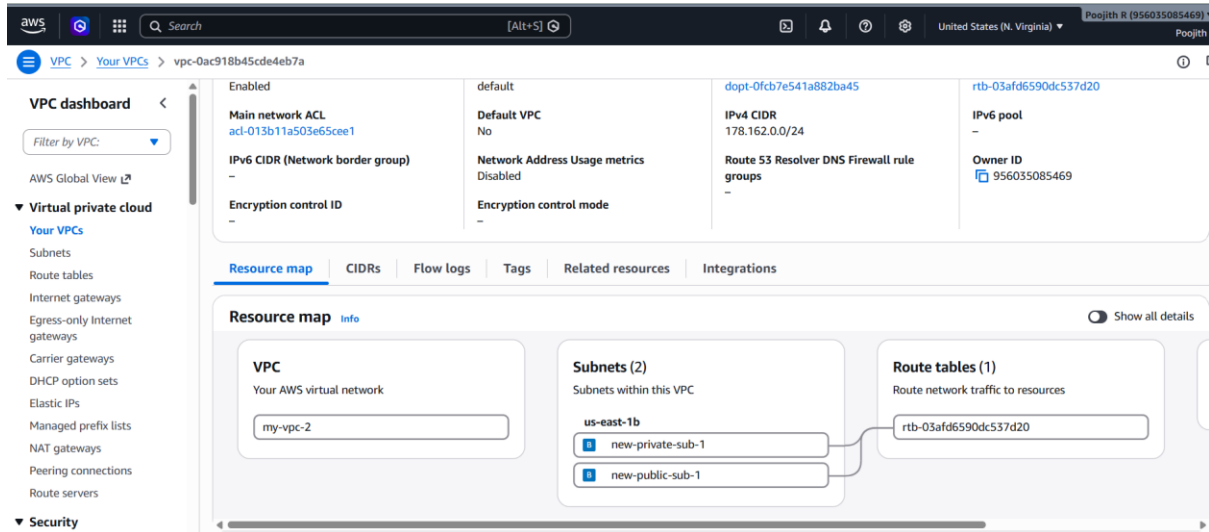


1. Create one VPC, with 1 public subnet and 1 private subnet.

➔ Created a VPC with a suitable CIDR block.

➔ Created one public subnet and one private subnet within the VPC.



2. Enable VPC peering for cross-region.

➔ Created a VPC peering connection between VPCs located in different AWS Regions.

➔ Specified the requester and acceptor VPCs with their respective Regions.

➔ Accepted the VPC peering request from the destination Region.

➔ Added routes to the respective route tables using the VPC peering connection.

➔ Updated the security groups to allow communication between the VPC CIDR ranges.

➔ Tested connectivity between resources in the cross-region VPCs

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VPC > Peering connections > pcx-0c6a82df1a4775955

VPC dashboard Filter by VPC: AWS Global View

Virtual private cloud

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- Elastic IPs
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- NAT gateways
- Peering connections**
- Route servers

Security

- Network ACLs
- Security groups

pcx-0c6a82df1a4775955 / Peering-1 Actions

Details Info

Requester owner ID 956035085469

Peering connection ID pcx-0c6a82df1a4775955

Status Initiating Request to 956035085469

Expiration time Tuesday, 8 September 2026 at 16:28:24 GMT+5:30

Accepter owner ID 956035085469

Requester VPC vpc-0f71a17bc2c6c9f4f / my-newVPC-1

Requester CIDRs 174.168.0.0/24

Requester Region Ohio (us-east-2)

VPC Peering connection ARN arn:aws:ec2:us-east-2:956035085469:vpc-peering-connection/pcx-0c6a82df1a4775955

Accepter VPC vpc-0ac918b45cde4eb7a

Accepter CIDRs -

Accepter Region N. Virginia (us-east-1)

DNS Route tables Tags

DNS settings Edit DNS settings

Requester VPC (vpc-0f71a17bc2c6c9f4f / my-newVPC-1) Info

Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses

Disabled

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VPC > Peering connections

VPC dashboard Filter by VPC: AWS Global View

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- Route servers

Security

- Network ACLs

Peering connections (1/1) Info Actions Create peering connection

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
-	pcx-0c6a82df1a4775955	Pending acceptance	vpc-0f71a17bc2c6c9f4f	vpc-0ac918b45cde4eb7a / my-...

pcx-0c6a82df1a4775955

Details DNS Route tables Tags

Details

Requester owner ID Accepter owner ID VPC Peering connection ARN

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#PeeringConnections:

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VPC > Peering connections

VPC dashboard Filter by VPC: AWS Global View

Virtual private cloud

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- Route servers

Security

- Network ACLs

Peering connections (1/1) Info Actions Create peering connection

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
-	pcx-0c6a82df1a4775955	Pending acceptance	vpc-0f71a17bc2c6c9f4f	vpc-0ac918b45cde4eb7a / my-...

Accept VPC peering connection request Info

Are you sure you want to accept this VPC peering connection request? (pcx-0c6a82df1a4775955)

Requester VPC vpc-0f71a17bc2c6c9f4f

Accepter VPC vpc-0ac918b45cde4eb7a / my-vpc-2

Requester CIDRs 174.168.0.0/24

Accepter CIDRs -

Requester Region Ohio (us-east-2)

Accepter Region N. Virginia (us-east-1)

Requester owner ID 956035085469(This account)

Accepter owner ID 956035085469(This account)

Cancel Accept request

pcx-0c6a82df1a4775955

Details DNS Route tables Tags

Details

Requester owner ID Accepter owner ID VPC Peering connection ARN

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VPC > Peering connections

Virtual private cloud

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- Route servers

Security

- Network ACLs

Peering connections (1) [Info](#) [Actions](#) [Create peering connection](#)

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
-	pcx-0c6a82df1a4775955	Provisioning	vpc-0f71a17bc2c6c9f4f	vpc-0ac918b45cde4eb7a / my-...

Select a peering connection above

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VPC > Route tables > rtb-0440908cd857bd2fe

Virtual private cloud

- Your VPCs
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- Route tables**
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

Updated routes for rtb-0440908cd857bd2fe / new-rt-1 successfully [Details](#)

rtb-0440908cd857bd2fe / new-rt-1 [Actions](#)

Details [Info](#)

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-0440908cd857bd2fe	No	-	-

VPC: [vpc-0f71a17bc2c6c9f4f | my-newVPC-1](#) Owner ID: [956035085469](#)

[Routes](#) Subnet associations Edge associations Route propagation Tags Related resources

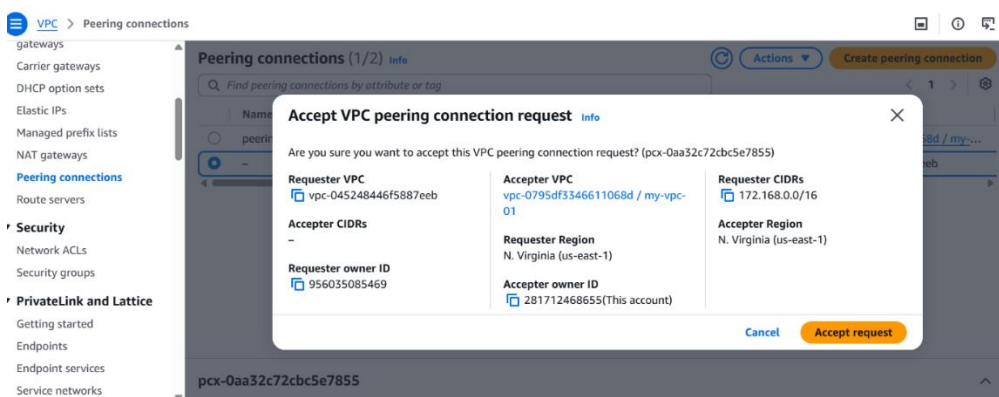
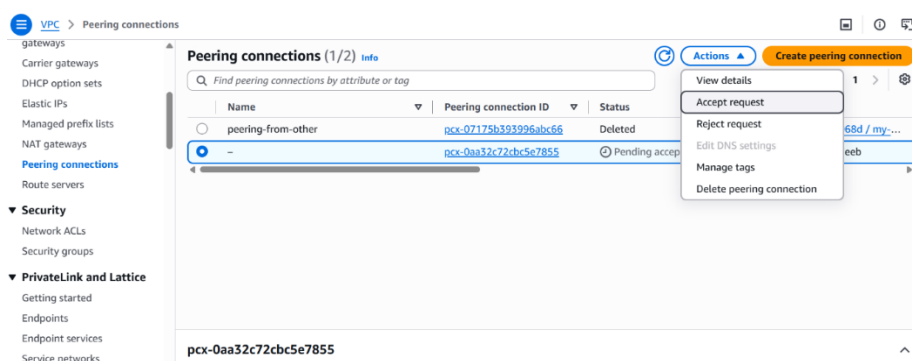
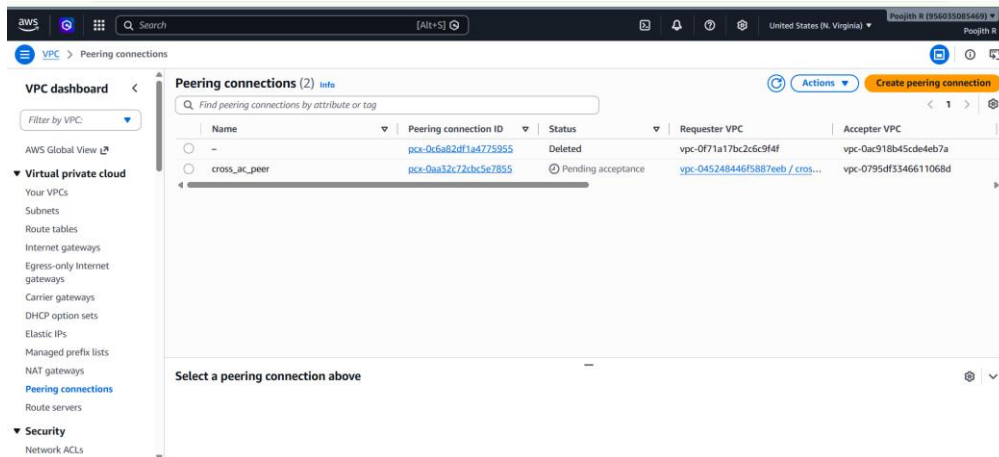
Routes (2) [Filter routes](#) [Both](#) [Edit routes](#)

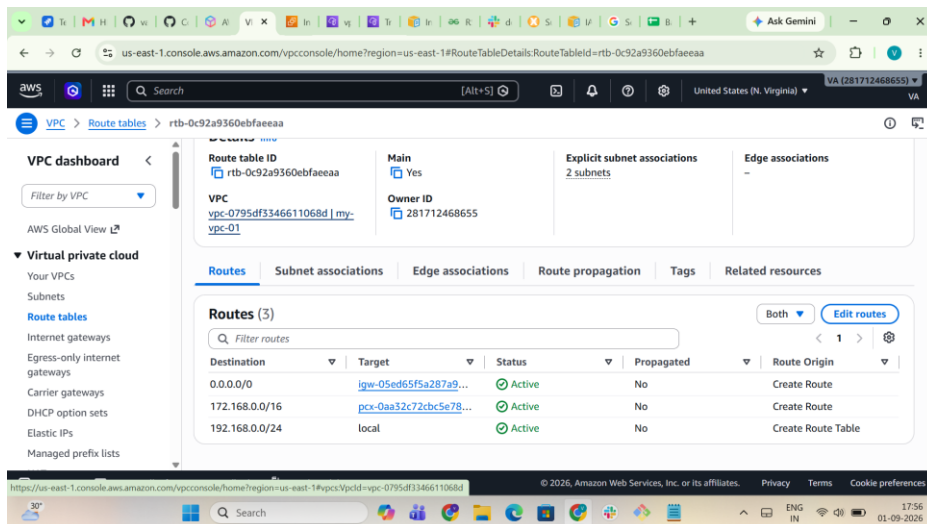
Destination	Target	Status	Propagated	Route Origin
174.168.0.0/24	local	Active	No	Create Route Table
178.162.0.0/24	pcx-0c6a82df1a4775955	Active	No	Create Route

```
root@ip-178-162-0-29:~# ping 174.168.0.10
PING 174.168.0.10 (174.168.0.10) 56(84) bytes of data:
64 bytes from 174.168.0.10: icmp_seq=1 ttl=127 time=11.2 ms
64 bytes from 174.168.0.10: icmp_seq=2 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=3 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=4 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=5 ttl=127 time=11.2 ms
64 bytes from 174.168.0.10: icmp_seq=6 ttl=127 time=11.2 ms
64 bytes from 174.168.0.10: icmp_seq=7 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=8 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=9 ttl=127 time=11.2 ms
64 bytes from 174.168.0.10: icmp_seq=10 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=11 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=12 ttl=127 time=11.3 ms
64 bytes from 174.168.0.10: icmp_seq=13 ttl=127 time=11.2 ms
64 bytes from 174.168.0.10: icmp_seq=14 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=15 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=16 ttl=127 time=11.2 ms
64 bytes from 174.168.0.10: icmp_seq=17 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=18 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=19 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=20 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=21 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=22 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=23 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=24 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=25 ttl=127 time=11.1 ms
64 bytes from 174.168.0.10: icmp_seq=26 ttl=127 time=11.1 ms
```

3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

- ➔ Created a VPC peering request from the VPC in the first AWS account.
- ➔ Specified the VPC ID and AWS account ID of the peer account.
- ➔ Accepted the VPC peering request from the second AWS account.
- ➔ Added routes to both VPC route tables using the VPC peering connection.
- ➔ Updated the security groups to allow traffic between the two VPCs.
- ➔ Verified connectivity between resources in the cross-account VPCs

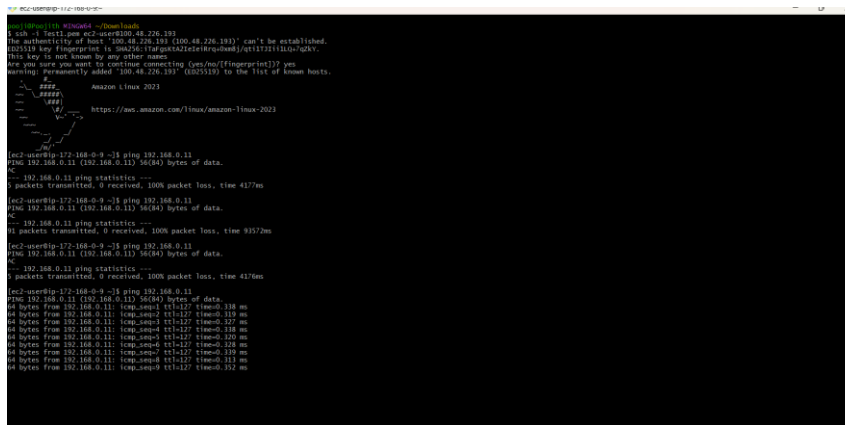




Peering connections (2) [Info](#)

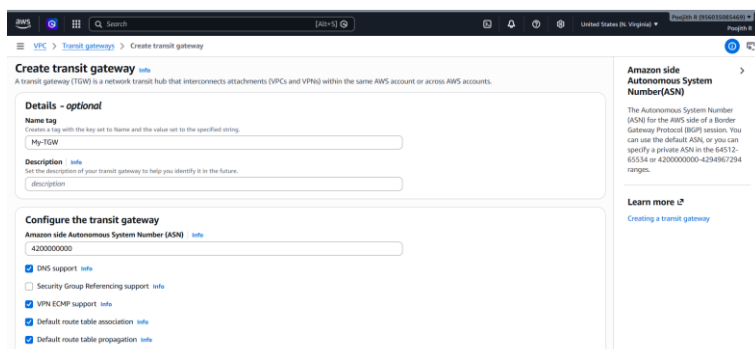
[Find peering connections by attribute or tag](#)

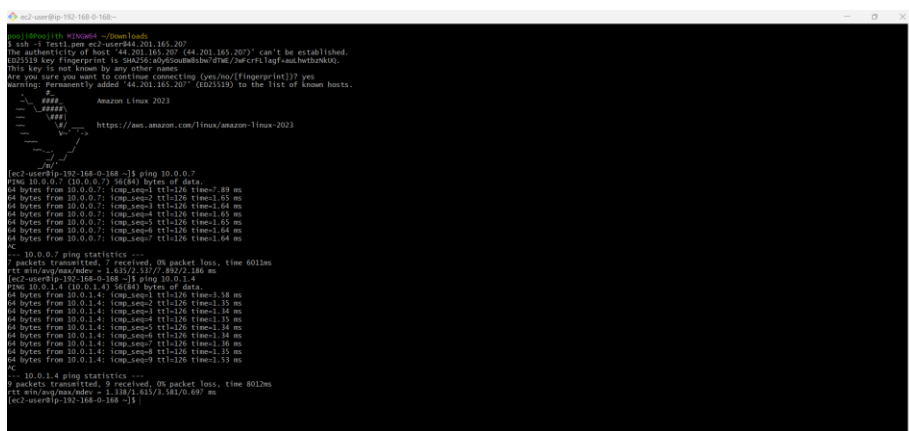
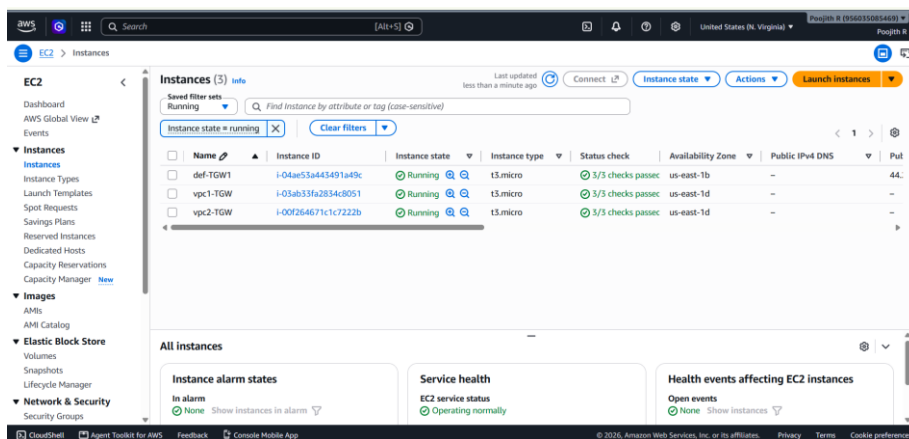
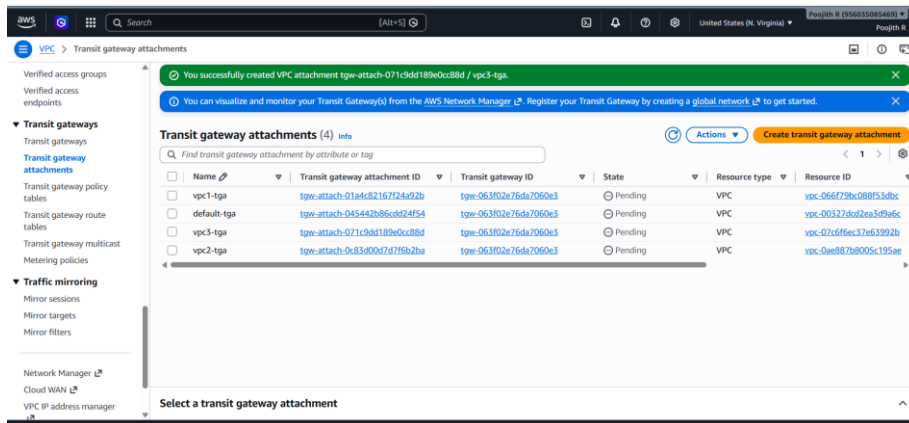
Name	Peering connection ID	Status	Requester VPC
-	pcx-0aa32c72cbc5e7855	Active	vpc-045248446f5887eeb



4. Set up a VPC Transit Gateway.

- ➔ Created a Transit Gateway in the AWS Region.
- ➔ Created a Transit Gateway attachment for the VPC's.
- ➔ Selected the required subnets for the Transit Gateway attachment.
- ➔ Associated the VPC attachment with the Transit Gateway route table.
- ➔ Added routes in the VPC route tables pointing to the Transit Gateway.
- ➔ Verified the Transit Gateway attachment and routing configuration.





5. Set up a VPC Endpoint.

- ➔ Created a VPC Endpoint for the required AWS service(s3).
- ➔ Selected the VPC where the endpoint was required.
- ➔ Selected the appropriate endpoint type (VPC) and service (s3).
- ➔ Associated the endpoint with the required route table/subnets.
- ➔ Configured the required security group and endpoint policies.
- ➔ Verified the endpoint status and tested connectivity to the AWS service

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EC2 > Instances

EC2

- Dashboard
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- Instance Types
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- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager
- Images**
- AMIs
- AMI Catalog
- Elastic Block Store**
- Volumes
- Snapshots
- Lifecycle Manager
- Network & Security**
- Security Groups

Instances (2) info

Find instance by attribute or tag (case-sensitive)

Save filter sets

Choose filter set

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Put
☐	EP1	i-0ac9138b995d47787	Running	t3.micro	Initializing	us-east-1a	-	98%
☐	EP2	i-062e4a839f5df395d	Running	t3.micro	Initializing	us-east-1a	-	-

All instances

Instance alarm states

In alarm

None Show instances in alarm

[View in CloudWatch](#)

Service health

EC2 service status

Operating normally

[View Service Health Dashboard](#)

Health events affecting EC2 instances

Open events

None Show instances

Upcoming events

None Show instances

[View AWS Health Dashboard](#)

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VPC > Route tables

VPC dashboard

Filter by VPC

AWS Global View

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- Managed prefix lists
- NAT gateways
- Route associations

Updated routes for rtb-0c45d8a50df83a672 / EP-rt-1 successfully

Details

Route tables (4) info

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-00827ef7bec1b22de	-	-	Yes	vpc-0615113cfbc93b4f End
☐	EP-rt-1	rtb-0c45d8a50df83a672	subnet-0ac73bhd7ee5ab...	-	No	vpc-0615113cfbc93b4f End
☐	EP-rt-2-prv	rtb-02407faa72630d55d	subnet-0e80424060148...	-	No	vpc-0615113cfbc93b4f End

Select a route table

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VPC > Subnets

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

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- Elastic IPs
- Managed prefix lists
- NAT gateways

Subnets (3) info

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-	subnet-08819a228ac33093	Available	vpc-0f3216a7937311169	Off	172.31.32.0/20
☐	EP-subnet-1	subnet-0ac73bhd7ee5ab6aa	Available	vpc-0615113cfbc93b4f Endp...	Off	10.0.0.0/28
☐	EP-subnet-2	subnet-0e80424060148ff80	Available	vpc-0615113cfbc93b4f Endp...	Off	10.0.0.16/28

Select a subnet

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VPC > Your VPCs

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

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- Managed prefix lists

Your VPCs

VPCs VPC encryption controls

Your VPCs (2) info

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv
☐	-	vpc-0f3216a7937311169	Available	-	-	Off	172
☐	Endpoint-test	vpc-0615113cfbc93b4f	Available	-	-	Off	101

Select a VPC above

